

Discrete Mathematics - Practical 5

Combinatorics, the basic counting rules: worked solutions
with the reasoning spelled out, step by step

Warm-up

This sheet is about *counting without listing*. Almost everything here rests on two rules. The **product rule**: if a thing is built by making a first choice in a ways and then, independently, a second choice in b ways, there are $a \cdot b$ ways in total (a grid with a rows and b columns). The **sum rule**: if a thing falls into separate, non-overlapping cases with a and b options, there are $a + b$ options in total. Two later problems add the **pigeonhole principle**: if you put $n + 1$ objects into n boxes, some box holds at least two. Counting is visual - draw the slots, the cases, or the boxes, and the formula reads itself off the picture. We solve sections 1 and 2 problems labelled 1.3, 1.5, 1.10, 2.5, 2.10, 2.12.

Problem 1.3 - Two books of different subjects

Problem 1.3

A shelf has 15 informatics books, 12 mathematics books, and 10 physics books (all distinct). In how many ways can you choose **two** books of **different** subjects?

The idea

"Different subjects" means the two books come from two different piles. So first decide *which two of the three subjects* the pair uses - there are three such subject-pairs: informatics+math, informatics+physics, math+physics. These three cases never overlap (a pair belongs to exactly one of them), so by the **sum rule** we add their counts. Inside one case, say informatics+math, the two picks are independent: any of the 15 informatics books with any of the 12 math books, which is $15 \cdot 12$ by the **product rule**. So the whole answer is one sum of three products.

Step 1 - Split into the three subject-pairs (sum rule). A valid pair uses exactly one of: {inf, math}, {inf, phys}, {math, phys}.

Step 2 - Count each pair (product rule).

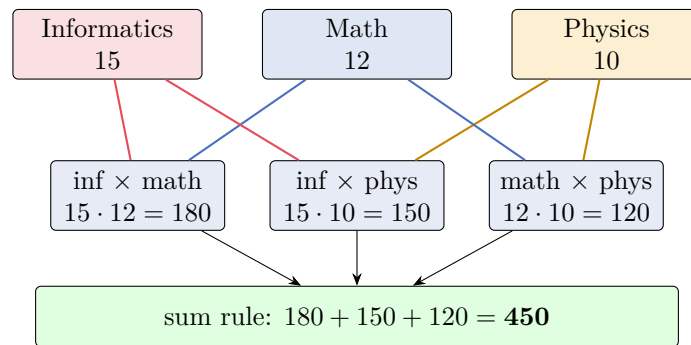
$$\underbrace{15 \cdot 12}_{\text{inf, math}} = 180, \quad \underbrace{15 \cdot 10}_{\text{inf, phys}} = 150, \quad \underbrace{12 \cdot 10}_{\text{math, phys}} = 120.$$

Step 3 - Add the cases.

$$180 + 150 + 120 = 450.$$

Always check. The pairs are unordered, but each product $15 \cdot 12$ already counts an *unordered* pair once: "an informatics book and a math book" is listed a single time, not twice, because the two books come from different labelled piles. No double counting, and the three cases are disjoint.
✓

$$\text{Answer: } 15 \cdot 12 + 15 \cdot 10 + 12 \cdot 10 = 180 + 150 + 120 = 450$$



★ Method to remember

Transferable technique: *multiply within a case, add across cases.* When objects split into disjoint types, count each type with the product rule, then sum the types (sum rule). Choosing two of three groups always gives exactly $\binom{3}{2} = 3$ cases.

Problem 1.5 - Numbers below 700 and divisibility

Problem 1.5

Among the natural numbers less than 700: how many are divisible by 5? Of those, how many are also divisible by 3? How many are divisible by both 3 and 5?

The idea

The multiples of d below a bound are evenly spaced: $d, 2d, 3d, \dots$. So "how many multiples of d are at most N " is just $\lfloor N/d \rfloor$ - you fit as many full steps of size d into N as you can. "Less than 700" means $1, 2, \dots, 699$, so $N = 699$. The only subtlety is the last question: "divisible by both 3 and 5" is the same as "divisible by $\text{lcm}(3, 5) = 15$ ", because 3 and 5 are coprime. And "divisible by 5 and also by 3" is the very same condition - so the second and third answers coincide.

Step 1 - Fix the range. Natural numbers less than 700 are $1, 2, \dots, 699$, so we use $N = 699$.

Step 2 - Divisible by 5.

$$\left\lfloor \frac{699}{5} \right\rfloor = 139 \quad (\text{these are } 5, 10, 15, \dots, 695).$$

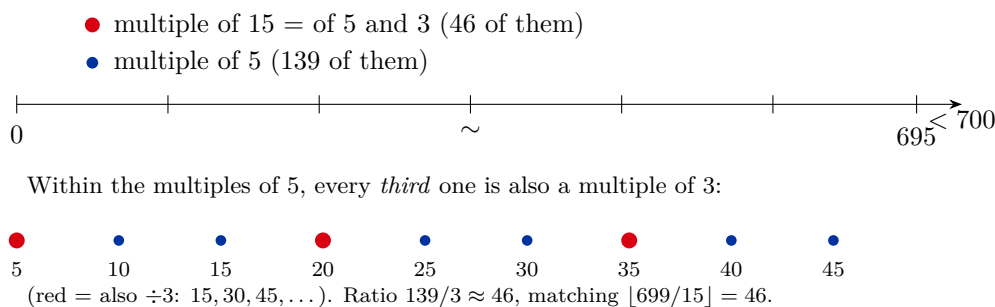
Step 3 - Of those, also divisible by 3. A number divisible by 5 and by 3 is divisible by $\text{lcm}(3, 5) = 15$ (true because $\text{gcd}(3, 5) = 1$). So count multiples of 15:

$$\left\lfloor \frac{699}{15} \right\rfloor = 46 \quad (\text{these are } 15, 30, \dots, 690).$$

Step 4 - Divisible by both 3 and 5. This is literally the same condition - divisible by 15 - so the count is again 46.

Always check. $695 = 5 \cdot 139 \leq 699$ and the next multiple $700 > 699$, so 139 is right. $690 = 15 \cdot 46 \leq 699$ and $15 \cdot 47 = 705 > 699$, so 46 is right. ✓

Answer: $\lfloor 699/5 \rfloor = 139$ divisible by 5; $\lfloor 699/15 \rfloor = 46$ of them by 3; **46** by both 3 and 5.



★ Method to remember

Transferable technique: the count of multiples of d in $\{1, \dots, N\}$ is $\lfloor N/d \rfloor$. "Divisible by a and by b " collapses to "divisible by $\text{lcm}(a, b)$ "; for coprime a, b that is just $a \cdot b$.

Problem 1.10 - Seven-digit numbers with a parity pattern

Problem 1.10

How many 7-digit numbers have an **odd** digit in every **even** position and an **even** digit in every **odd** position? (Position 1 is the leading digit and cannot be 0.)

The idea

A 7-digit number is just seven slots filled left to right, and each slot's digit is chosen *independently* of the others, so this is a pure product-rule count: multiply the number of allowed digits in each slot. Read the rule slot by slot. Odd positions 1, 3, 5, 7 want an *even* digit, of which there are five: $\{0, 2, 4, 6, 8\}$. Even positions 2, 4, 6 want an *odd* digit, also five: $\{1, 3, 5, 7, 9\}$. The single catch is the very first slot: it is an odd position, so it wants an even digit, but it may not be 0 (else the number is not really 7-digit). That removes exactly one option, leaving $\{2, 4, 6, 8\}$, i.e. 4 choices. Every other slot keeps its full 5.

Step 1 - Classify the slots. Positions 1 to 7. *Odd* positions $\{1, 3, 5, 7\}$ take an even digit; *even* positions $\{2, 4, 6\}$ take an odd digit.

Step 2 - Count the choices per slot. Even digits $\{0, 2, 4, 6, 8\}$: 5 each. Odd digits $\{1, 3, 5, 7, 9\}$: 5 each. Position 1 is special: even digit but not 0, so $\{2, 4, 6, 8\}$, only 4 choices.

Step 3 - Multiply (product rule). One slot with 4 choices (position 1), and the other six slots with 5 choices each:

$$\underbrace{4}_{\text{pos 1}} \cdot \underbrace{5 \cdot 5 \cdot 5}_{\text{pos 3,5,7}} \cdot \underbrace{5 \cdot 5 \cdot 5}_{\text{pos 2,4,6}} = 4 \cdot 5^6 = 4 \cdot 15625 = 62500.$$

Always check. If position 1 could also be 0 we would get $5^7 = 78125$; the ban on a leading 0 removes exactly the numbers with 0 in slot 1, which is a $\frac{1}{5}$ share, leaving $\frac{4}{5} \cdot 5^7 = 4 \cdot 5^6 = 62500$. Same number, reached a second way. ✓

Answer: $4 \cdot 5^6 = 62500$

pos 1	pos 2	pos 3	pos 4	pos 5	pos 6	pos 7
even, not 0 4	odd 5	even 5	odd 5	even 5	odd 5	even 5
{2, 4, 6, 8}	{1, 3, 5, 7, 9}	{0, 2, 4, 6, 8}	{1, 3, 5, 7, 9}	{0, 2, 4, 6, 8}	{1, 3, 5, 7, 9}	{0, 2, 4, 6, 8}

$$4 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5 = 4 \cdot 5^6 = \mathbf{62500}$$

blue = odd position (even digit), orange = even position (odd digit), red = leading slot

★ Method to remember

Transferable technique: for "fill the slots" problems where each position is chosen independently, multiply the per-slot option counts. Handle the leading-digit "no 0" restriction by reducing that one slot's count.

Problem 2.5 - Six boys and seven girls in a row, ends are boys

Problem 2.5

Six boys and seven girls (13 distinct people) stand in a row of 13 places. In how many ways can they line up so that the **first and the last** place are both occupied by boys?

The idea

Order matters (a row), so this is an arrangement count. The clean move is to *seat the constrained spots first*: place the two end positions, which are forced to be boys, before anything else. Choose the boy on the left end (6 choices), then the boy on the right end (5 choices left), so $6 \cdot 5$ ways for the two ends. Now the two ends are settled and the restriction is fully used up: the remaining 11 people (the other 4 boys and all 7 girls) are completely free to fill the middle 11 places in any order, which is $11!$ ways. Multiply the independent stages.

Step 1 - Fill the ends (the only restriction). Left end: any of the 6 boys. Right end: any of the remaining 5 boys. That is $6 \cdot 5 = 30$ ordered ways - this is the arrangement $A_6^2 = 6 \cdot 5$.

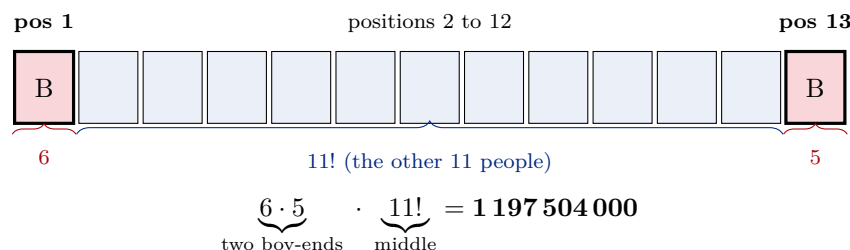
Step 2 - Fill the middle freely. After the ends are fixed, 11 distinct people remain for 11 middle places: $11!$ orderings.

Step 3 - Multiply (product rule).

$$6 \cdot 5 \cdot 11! = 30 \cdot 39\,916\,800 = 1\,197\,504\,000.$$

Always check. Total arrangements with no restriction would be $13!$. Our condition forces both ends to be among the 6 boys; the fraction of all $13!$ lineups with a boy at each end is $\frac{6}{13} \cdot \frac{5}{12}$, giving $13! \cdot \frac{6 \cdot 5}{13 \cdot 12} = \frac{13!}{156} \cdot 30$. Since $13! = 6\,227\,020\,800$, this is $6\,227\,020\,800 \cdot \frac{30}{156} = 1\,197\,504\,000$. Same number. ✓

Answer: $6 \cdot 5 \cdot 11! = 30 \cdot 11! = \mathbf{1\,197\,504\,000}$



★ Method to remember

Transferable technique: *place the restricted positions first*, then arrange the rest freely. Choosing ordered people for k named spots out of n is $A_n^k = n(n-1) \cdots (n-k+1)$; the leftover $n-k$ people fill the remaining spots in $(n-k)!$ ways.

Problem 2.10 - Three mutual friends or three mutual strangers

Problem 2.10

In a room of 6 people, every two of them are either **friends** or **strangers**. Prove that there are always three people who are pairwise friends, or three people who are pairwise strangers.

The idea

Model it as a complete graph K_6 : 6 dots, one edge between every pair, each edge coloured **red** (friends) or **blue** (strangers). The claim is that some triangle is monochromatic. The trick is to look from *one* person P . P has 5 edges; two colours; by the **pigeonhole principle** at least $\lceil 5/2 \rceil = 3$ of them share a colour - say P is friends (red) with three people A, B, C . Now everything hinges on the triangle A, B, C . If even one of the pairs AB, BC, CA is red, that pair plus P is an all-red triangle (three mutual friends). If *none* of them is red, then AB, BC, CA are all blue, so A, B, C themselves are an all-blue triangle (three mutual strangers). There is no escape. This is exactly the statement $R(3, 3) \leq 6$.

Step 1 - Set up the model. Vertices = the 6 people; colour each of the $\binom{6}{2} = 15$ edges red (friends) or blue (strangers). "Three mutual friends" = a red triangle; "three mutual strangers" = a blue triangle.

Step 2 - Pigeonhole at one vertex. Fix any person P . The 5 edges leaving P are coloured with 2 colours, and $5 > 2 \cdot 2$, so by pigeonhole at least 3 of them have the same colour. Without loss of generality, P is joined in **red** to three people A, B, C (if instead the majority colour is blue, swap the words "friends"/"strangers" throughout - the argument is identical).

Step 3 - Examine the triangle A, B, C (two cases).

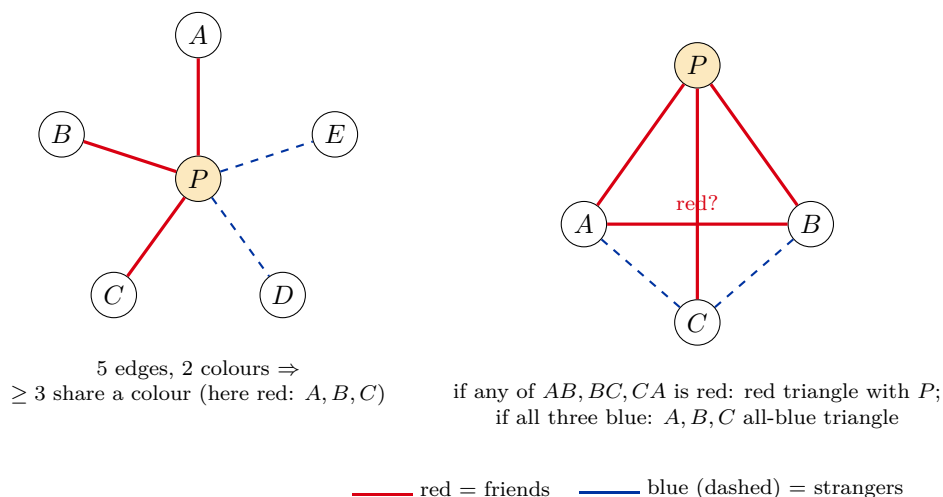
- *Some edge among A, B, C is red*, say AB . Then PA, PB, AB are all red: P, A, B are three mutual friends. Done.
- *No edge among A, B, C is red*. Then AB, BC, CA are all blue, so A, B, C are three mutual strangers. Done.

One of the two cases must occur, so a monochromatic triangle always exists. ■

Always check (is 6 the right number?). With only 5 people the claim can fail: colour K_5 as a red 5-cycle and a blue 5-cycle (the "pentagon and pentagram") and no triangle is monochromatic. So 5 is not enough and 6 is sharp; that is why the problem says 6. (Brute force confirms: among

all 2^{15} colourings of K_6 , zero avoid a monochromatic triangle, while K_5 has colourings that do.)
✓

Answer: Such a monochromatic triangle always exists: this is $R(3, 3) \leq 6$.



★ Method to remember

Transferable technique (Ramsey via pigeonhole): to force structure, focus on one vertex, pigeonhole its edges into a majority colour, then check the small group that majority creates. The two-case split "some edge inside is colour X " vs "none is" is the engine. This proves $R(3, 3) \leq 6$.

Problem 2.12 - One of $7, 77, 777, \dots$ is divisible by 2013

Problem 2.12

Consider the sequence $7, 77, 777, 7777, \dots$ (term n is the number written with n sevens). Prove that among its first 2014 terms, at least one is divisible by 2013.

The idea

You cannot point to which term works without computing, but you do not need to - you only need to prove one *exists*. The tool is the **pigeonhole principle** applied to remainders. Look at the first 2014 terms modulo 2013. A remainder mod 2013 is one of only 2013 values $\{0, 1, \dots, 2012\}$ - that is 2013 "boxes". But we have 2014 terms to drop into them, so two terms must land in the same box, i.e. share a remainder. Their *difference* is then divisible by 2013. The last step is to see that this difference is itself a number of the right shape (sevens followed by zeros), and the zeros (a power of 10) can be stripped off because 2013 is coprime to 10. What remains is a term of the sequence divisible by 2013.

Step 1 - Set up boxes and objects. Let $a_n = \underbrace{77 \dots 7}_{n \text{ sevens}}$. Look at the remainders $a_1 \bmod 2013$, $a_2 \bmod 2013$, $\dots$, $a_{2014} \bmod 2013$. Each remainder is one of the 2013 values $0, 1, \dots, 2012$.

Step 2 - Pigeonhole. We have 2014 remainders but only 2013 possible values, so two of them are equal:

$$a_i \equiv a_j \pmod{2013}, \quad \text{for some } 1 \leq i < j \leq 2014.$$

(If instead one of the remainders is already 0, that term is divisible by 2013 and we are done immediately.)

Step 3 - The difference is divisible by 2013. From $a_i \equiv a_j$ we get $2013 \mid (a_j - a_i)$. Subtracting the two repunits-of-sevens:

$$a_j - a_i = \underbrace{77 \cdots 7}_j - \underbrace{77 \cdots 7}_i = \underbrace{77 \cdots 7}_{j-i} \underbrace{00 \cdots 0}_i = a_{j-i} \cdot 10^i.$$

(The top $j - i$ sevens survive; the bottom i digits cancel to zeros.)

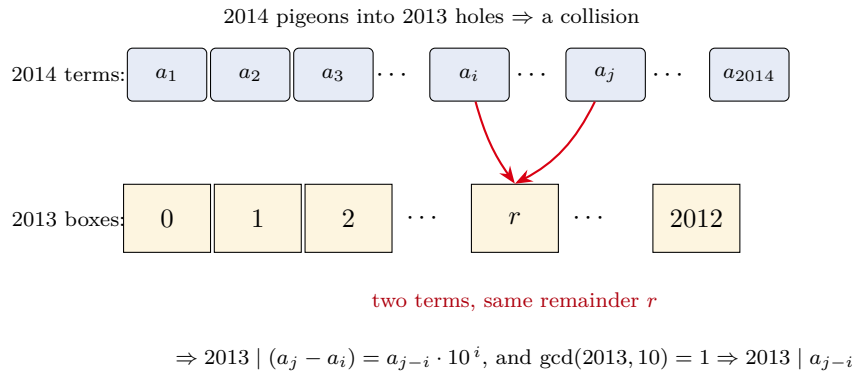
Step 4 - Strip the power of 10. $2013 = 3 \cdot 11 \cdot 61$, which shares no factor with $10 = 2 \cdot 5$, so $\gcd(2013, 10^i) = 1$. Since $2013 \mid a_{j-i} \cdot 10^i$ and 2013 is coprime to 10^i , it must divide the other factor:

$$2013 \mid a_{j-i}.$$

Finally $1 \leq j - i \leq 2013 < 2014$, so a_{j-i} is one of the first 2014 terms. ■

Always check. The proof only guarantees *existence*; a direct computation shows the first term that actually works is a_{60} (the number with sixty 7s), and indeed $60 \leq 2014$. So the existence claim is not just true but comfortably so. ✓

Answer: Some a_{j-i} with $1 \leq j - i \leq 2013$ is divisible by 2013; in fact a_{60} already is.



★ Method to remember

Transferable technique: to show "some term is divisible by m ," take $m + 1$ candidates and pigeonhole their remainders mod m ; a repeat gives a difference divisible by m . If the difference carries a factor 10^k , drop it using $\gcd(m, 10) = 1$.